SYS 304
Trade Studies, Risk, and Decision Analysis

Advanced Trade Studies and Decision Analysis Methods

Simulation Approaches 1

Spreadsheet Simulation

- Popular, very used tool.
- Can be used for simple simulation models.
  - Typically, static models.
    - Risk analysis, financial/investment scenarios.
  - Only the simplest of dynamic models.
- Two examples:
  - Newsvendor problem.
  - Waiting times in single-server queue.
    - Special recursion valid only in this case.

Newsvendor Problem – Setup

- Newsvendor sells newspapers on the street.
  - Buys for c = \$0.55 each, sells for r = \$1.00 each.
- Each morning, buys q copies.
  - q is a fixed number, same every day.
- Demand during a day: $D = \max(\lfloor X \rceil, 0)$.
  - $X \sim$ normal ($\mu = 135.7$, $\sigma = 27.1$), from historical data.
    - $\lfloor X \rceil$ rounds X to nearest integer.
- If $D \leq q$, satisfy all demand, and $q - D \geq 0$ left over, sell for scrap at s = \$0.03 each.
- If $D > q$, sells out (sells all q copies), no scrap.
  - But missed out on $D - q > 0$ sales.
- What should q be?

Newsvendor Problem – Formulation

- Choose q to maximize expected profit per day.
  - q too small – sell out, miss \$0.45 profit per paper.
  - q too big – have left over, scrap at a loss of \$0.52 per paper.
- Classic operations-research problem.
  - Many versions, variants, extensions, applications.
  - Much research on exact solution in certain cases.
  - But easy to simulate, even in a spreadsheet.
- Profit in a day, as a function of q:

$$W(q) = \underbrace{r \min(D, q)}_{\text{Sales revenue}} + \underbrace{s \max(q - D, 0)}_{\text{Scrap revenue}} - \underbrace{cq}_{\text{Cost}}.$$

  - $W(q)$ is a random variable – profit varies from day to day.
- Maximize $E(W(q))$ over nonnegative integers q.

Newsvendor Problem – Simulation

- Set trial value of q, generate demand D, compute profit for that day.
  - Then repeat this for many days independently, average to estimate E(W(q)).
    - Also get confidence interval, estimate of P(loss), histogram of $W(q)$.
  - Try for a range of values of q.
- Need to generate demand $D = \max(\lfloor X \rceil, 0)$.
  - So need to generate $X \sim$ normal ($\mu = 135.7$, $\sigma = 27.1$).
    - (Much) ahead - generating random variates.
  - In this case, generate $X = \Phi^{-1}{}_{\mu,\sigma}(U)$.

    U is a random number distributed uniformly on [0, 1].

    $\Phi^{-1}{}_{\mu,\sigma}(U)$ is cumulative distribution function of normal (μ, σ) distribution.

Newsvendor Problem – Excel

- File Newsvendor.xls
- Input parameters in cells B4 – B8 (blue).
- Trial values for *q* in row 2 (pink).
- Day number (1, 2, ..., 30) in column D.
- Demands in column E for each day:

$$= \texttt{MAX(}\underbrace{\texttt{ROUND}}_{\text{Rounding function}}\texttt{(}\underbrace{\underbrace{\texttt{NORMINV}}_{\Phi^{-1}}\texttt{(}\underbrace{\texttt{RAND()}}_{U(0,\ 1)\text{ random number}}\texttt{, }\underbrace{\texttt{\$B\$7}}_{\mu}\texttt{, }\underbrace{\texttt{\$B\$8}}_{\sigma}\texttt{)}}_{X \sim \text{normal }(\mu,\ \sigma)}\texttt{, }\underbrace{\texttt{0}}_{\text{Round to nearest integer}}\texttt{), }\underbrace{\texttt{0}}_{\text{MAX 2nd argument}}\texttt{)}$$

`RAND()` *is "volatile" so regenerates on any edit, or F9 key*

$ *pins down following column or row when copying*

Newsvendor Problem – Excel (cont'd.)

- For each q:
  - "Sold" column: number of papers sold that day.
  - "Scrap" column: number of papers scrapped that day.
  - "Profit" column: profit (+, –, 0) that day.
  - Placement of "`$`" in formulas to facilitate copying.
- At bottom of "Profit" columns (green):
  - Average profit over 30 days.
  - Half-width of 95% confidence interval on E(W(q)).
    - Value 2.045 is upper 0.975 critical point of t distribution with 29 d.f.
    - Plot confidence intervals as "I-beams" on left edge.
  - Estimate of P(W(q) < 0).
    - Uses **`COUNTIF`** function.
- Histograms of W(q) at bottom.
  - Vertical red line at 0, separates profits, losses.

Newsvendor Problem – Results

- Fine point – used same daily demands (column E) for each day, across all trial values of *q*.
  - Would have been valid to generate them independently.
  - Why is it better to use the same demands for all *q*?
- Results:
  - Best *q* is about 140, maybe a little less.
  - Randomness in all the results (tap F9 key).
    - All demands, profits, graphics change.
    - Confidence-interval, histogram plots change.
    - Reminder that these are random outputs, random plots.
  - Higher *q* $\Rightarrow$ more variability in profit.
    - Histograms at bottom are wider for larger *q*.
    - Higher chance of both large profits, but higher chance of loss, too.
    - Risk/return tradeoff can be quantified – risk taker vs. risk-averse.

Single-Server Queue – Setup

- Like hand simulation, but:
  - Interarrival times ~ exponential with mean $1/\lambda = 1.6$ min.
  - Service times ~ uniform on $[a, b] = [0.27, 2.29]$ min.
  - Stop when 50th waiting time in queue is observed.
    - i.e., when 50th customer *begins* service, not exits system.
- Watch waiting times in queue $WQ_1, WQ_2, \ldots, WQ_{50}$.
  - Important – not watching anything else, unlike before.
- S_i = service time of customer i.
- A_i = interarrival time between custs. i and $i+1$.
- *Lindley's recursion* (1952): Initialize $WQ_1 = 0$, $Wq_{i+1} = \max(WQ_i + S_i - A_i, 0)$, $i = 2, 3, \ldots$

Single-Server Queue – Simulation

- Need to generate random variates: let $U \sim$ U[0, 1].
  - Exponential (mean $1/\lambda$): $A_i = -(1/\lambda) \ln(1 - U)$
  - Uniform on $[a, b]$: $S_i = a + (b - a) U$
- File `MU1.xls`
- Input parameters in cells B4 – B6 (blue).
  - Some theoretical outputs in cells B8 – B10.
- Customer number (i = 1, 2, ..., 50) in column D.
- Five IID replications (three columns for each).
  - IA = interarrival times, S = service times.
  - WQ = waiting times in queue (plot, thin curves).
    - First one initialized to 0, remainder use Lindley's recursion.
      - Curves rise from 0, variation increases toward right.
    - Creates positive autocorrelation down the WQ columns.
      - Curves have less abrupt jumps than if WQ_i's were independent.

Single-Server Queue – Results

- Column averages (green).
  - Average interarrival, service times close to expectations.
  - Average WQ_i within each replication.
    - Not too far from steady-state expectation.
    - Considerable variation.
    - Many are below it (why?).

$$E(WQ_\infty) = \lambda \frac{Var(S_i) + (E(S_i))^2}{2(1 - \lambda E(S_i))}$$

- Cross-replication (by customer) averages.
  - Column T, thick line in plot to dampen noise.
- Why no sample variance, histograms of WQ_i's?
  - Could have computed both, as in newsvendor.
  - Nonstationarity – what is a "typical" WQ_i here?
  - Autocorrelation – biases variance estimate, may bias histogram if run is not "long enough".

Spreadsheet Simulation Software

- Popular for static models.
  - Add-ins to Excel: @RISK, Crystal Ball, RiskSim (TreePlan).
- Inadequate tool for dynamic simulations if there’s any complexity.
  - Extremely easy to simulate the single-server queue in Arena.
  - Can build very complex dynamic models with Arena.